

In analyses of genetic predictors for health outcomes such as disease status, we often employ population-based designs such as the case-control study. A flaw of this design, however, is its vulnerability to population stratification, which motivates the use of family-based designs, such as case-parents studies where cases and their parents are genotyped (forming a “trio”), allowing one’s parents to be considered as controls. Family-based studies still suffer from unaddressed confounders and factors which may impact potential inferences. One such factor is maternal genetic factors, which are relevant when studying diseases known to be dependent on one’s gestational environment such as spina bifida (Hobbs et al., 2014). Maternal gene-environment interactions, such as the extent smoking impacts a mother’s child in her womb being dependent on the mother’s genotype, are another potential confounder. We also have mating asymmetry (MA), where mothers and fathers do not have the same distribution of genetic factors among the population, or $\Pr(M = g_1, F = g_2) \neq \Pr(M = g_2, F = g_1)$ where M and F are the genotypes of the mother and father respectively, and g_1 and g_2 are two distinct genetic factors. This in turn can cause us to identify false positives such as maternal effects being present should we assume the two distributions are the same when it is really due to population structures. Although this subject has not been thoroughly studied, Bourgey et al. (2011) have shown that MA can be observed in human populations, to the extent of creating significant effects in models that do not consider MA. Lastly, we have parent-of-origin effects (imprinting), where a child’s outcome is dependent not only on their genotype but also on the parent who contributed the genetic factor of interest. In terms of a probability model, we would include I_m and I_p as imprinting parameters, where the probability of disease is multiplied by I_m if they received a high-risk genetic factor maternally, and multiplied by I_p if received paternally (Ainsworth et al., 2011).

Weinberg et al. (1998) proposed a log-linear model for case-family data allowing one to draw inferences such as the relative risk of a disease given some genetic factor, which was further extended to a hybrid design, where trio data of cases and their biological parents could be compared to parents of controls (Weinberg and Umbach, 2005). Although no official software for their model exists, one could have used a now deprecated software called LEM, which also had methods for missing data (Vermunt, 1997). Other software for modelling case-family data includes Howey and Cordell (2012)’s EMIM, which was similar but used a multinomial model, and Gjerdevik et al. (2019)’s Haplin, which employed a log-linear model. As models for case-family data, all of them could account for maternal genetic factors and imprinting, however, a limitation of this family-based data was that no inferences about environmental effects or mating asymmetry could be obtained, motivating the use of the hybrid model. With this model, one could include broad confounders, such as environmental effects like smoking, while still considering all of the previously mentioned cofactors. Unfortunately, no software methods for this model are currently available, and so users must implement it themselves to perform this analysis.

In our research project, we will create a freely available R package allowing anyone to analyze trio data with the hybrid model. This includes the basic code to fit the model as a generalized linear model (GLM) object in R, which will allow users to construct confidence intervals and perform significance tests on covariates using functions in base R. The model will be designed to be robust, allowing users to input data with or without controls and account for missing data using the Expectation-Maximization (EM) algorithm. We will also conduct several simulation studies to assess the model’s accuracy and power in simulated datasets, which could help elucidate the strengths and weaknesses of the model. As a GLM, the model in our package can have a wide variety of cofactors added to it by advanced users. By default, our package will consider the common confounders we described before as covariates in the model, with users having the option to disable them.

With this package, our goal is to allow anyone with the appropriate trio data to perform an extensive analysis of the potential contributors to disease. We expect that the inferences drawn from the model will help inform public health recommendations and for screening high-risk patients.

References

- Ainsworth, H. F., J. Unwin, D. L. Jamison, and H. J. Cordell (2011, January). Investigation of maternal effects, maternal-fetal interactions and parent-of-origin effects (imprinting), using mothers and their offspring. *Genet. Epidemiol.* 35(1), 19–45.
- Bourgey, M., J. Healy, P. Saint-Onge, H. Massé, D. Sinnett, and M.-H. Roy-Gagnon (2011). Genome-wide detection and characterization of mating asymmetry in human populations. *Genetic Epidemiology* 35(6), 526–535.
- Gjerdevik, M., A. Jugessur, Ø. A. Haaland, J. Romanowska, R. T. Lie, H. J. Cordell, and H. K. Gjessing (2019, April). Haplin power analysis: a software module for power and sample size calculations in genetic association analyses of family triads and unrelated controls. *BMC Bioinformatics* 20(1), 165.
- Hobbs, C. A., M. A. Cleves, S. L. Macleod, S. W. Erickson, X. Tang, J. Li, M. Li, T. Nick, S. Malik, and National Birth Defects Prevention Study (2014, February). Conotruncal heart defects and common variants in maternal and fetal genes in folate, homocysteine, and transsulfuration pathways. *Birth Defects Res. A Clin. Mol. Teratol.* 100(2), 116–126.
- Howey, R. and H. J. Cordell (2012, June). PREMIM and EMIM: tools for estimation of maternal, imprinting and interaction effects using multinomial modelling. *BMC Bioinformatics* 13(1), 149.
- Vermunt, J. (1997, January). Lem: A general program for the analysis of categorical data.
- Weinberg, C. R. and D. M. Umbach (2005, October). A hybrid design for studying genetic influences on risk of diseases with onset early in life. *Am. J. Hum. Genet.* 77(4), 627–636.
- Weinberg, C. R., A. J. Wilcox, and R. T. Lie (1998, April). A log-linear approach to case-parent-triad data: assessing effects of disease genes that act either directly or through maternal effects and that may be subject to parental imprinting. *Am. J. Hum. Genet.* 62(4), 969–978.